Math 534H - Spring 2026 - Lecture 3

Let's check that indeed for any function f of one variable $u(x,t) = f(x-ct)$ is a solution of $u_t + cu_x = 0$.

$\left(\begin{array}{c} x-ct = \text{const} \\ \text{characteristic} \\ \text{lines} \end{array} \right)$

$$u_x = f'(\underbrace{x-ct}) \cdot 1 = f'$$

Chain rule

$$1 = \frac{\partial (x-ct)}{\partial x}$$

$$u_t = f'(\underbrace{x-ct}) \cdot (-c) = f'(-c)$$

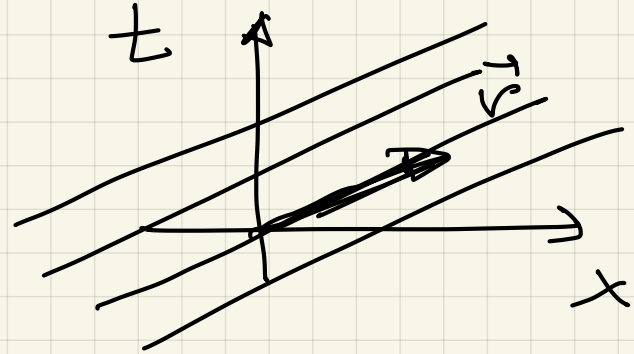
$$u_t + cu_x = f'(-c) + cf' = 0$$

Note this is true for all differentiable functions f of one variable.

concentration of dye/pollutant

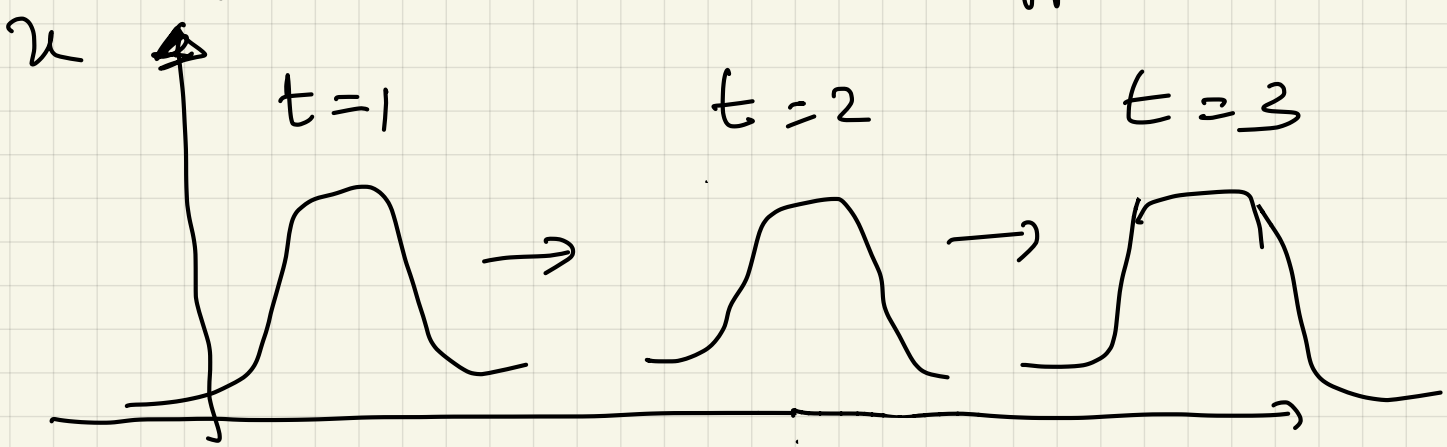
<u>Remark 1</u>: What $u(x,t) = f(x-ct)$ is that the substance (dye/pollutant) is transported to the right at speed c (= speed in which fluid is flowing).

In the xt-plane it moves precisely along the characteristic lines



$\vec{v} = (c, 1)$.

And if we look at what happens as "a move"



<u>Rmk 2</u> : If one looks at (x, t, u) space then the solution u defines a surface made up of parallel horizontal lines (lifted by f) looking like corrugated anodized aluminum or galvanized iron.

<u>Remark 3</u> : If we specify an initial value (auxiliary condition) for u then that will determine the f and hence a <u>particular</u> <u>solution</u> u.

<u>Example</u> : Solve $2u_y - 7u_x = 0$ subject to $u(x,0) = x^2$

The solution $u = u(x,y)$

$$\underbrace{-7}\, u_x + \underbrace{2}\, u_y = 0 .$$

$$\vec{v} = (\underbrace{-7}_{a}, \underbrace{2}_{b})$$

$$D_{\vec{v}}\, u = \nabla_{xy} u \cdot \vec{v} = 0$$

$a u_x + b u_y = 0$

$\vec{v} = (a, b)$

$\vec{v}^{\perp} = (-b, a)$

Then the <u>characteristic lines</u> note $\vec{v}^{\perp} = (2, 7)$

so the characteristic lines $\boxed{2x + 7y = \text{constant}}$

(" " " $\vec{v}^{\perp} \cdot (x,y) = \text{const}$).

<u>Aside</u>: this is the same as noting that the slope of $\vec{v}$ is $\frac{b}{a} = -\frac{2}{7}$ so characteristic lines are $\boxed{y = -\frac{2}{7}x + \text{const}}$

So the general solution will be $\boxed{u(x,y) = f(2x+7y)}$

But now we are told that $\underline{u(x,0) = x^2}$

That is $u(x,0) = \underline{f(2x) = x^2} \Longrightarrow$

$f(w) := \dfrac{w^2}{4} \longleftarrow$ determines f

$w \in \mathbb{R}$

Hence the $\boxed{\text{particular solution}}$ satisfying both <u>the equation</u> <u>and</u> <u>the initial condition</u> is

$$\boxed{u(x,y) = \frac{(2x+7y)^2}{4}}$$

Check that this is indeed a solution.

<u>IC</u>: $u(x,0) = \dfrac{(2x+7\cdot 0)^2}{4} = \dfrac{4x^2}{4} = x^2$ ✓

<u>Eq</u>: $-7u_x + 2u_y = 0$? $\qquad u(x,y) = \dfrac{(2x+7y)^2}{4}$

$$u_x = \frac{2(2x+7y)\cdot 2}{4} = 2x+7y$$

$$u_y = \frac{2(2x+7y)\cdot 7}{4} = \frac{7}{2}(2x+7y).$$

Hence $-7\underbrace{(2x+7y)}_{u_x} + \cancel{2}\underbrace{\left(\frac{7}{\cancel{2}}(2x+7y)\right)}_{u_y} = 0$ ✓

as desired.

Derivation of $\boxed{u_t + c\,u_x = 0}$ ①.

The amount of pollutant/dye in an interval $[0, x_0]$ in the x-axis at a <u>fixed time t</u> is

$$\int_0^{x_0} u(x,t)\,dx \quad \text{let's call it } M. \quad (= M(t))$$

At a later time say <u>$t+h$</u> $(h>0)$ the same particles (of pollutant/dye) have moved to the right by $c\cdot h$. Hence

$$M = \int_0^{x_0} u(x,t)\,dx = \int_{ch}^{x_0+ch} u(x,t+h)\,dx$$

So if we differentiate the equality above both sides with respect to x_0 we get (by FTC)

$$u(x_0, t) = u(x_0 + ch, t+h)$$

Next differentiate both sides with respect to h (and then we'll set $h=0$). We get

$$0 = u_x(x_0+ch, t+h) \cdot c + u_t(x_0+ch, t+h) \cdot 1$$

Now set $h=0$ to get

$$0 = u_x(x_0, t) \cdot c + u_t(x_0, t)$$

This is exactly equation 1

Another take at solving $\boxed{a u_x + b u_y} = 0$ using what's called "<u>Coordinate Method</u>" (using a <u>change of variables</u>): $(x, y) \to (\xi, \eta)$

$$\begin{cases} \xi = ax + by = (a, b)\cdot(x, y) = \vec{v}\cdot(x, y) \\ \eta = bx - ay = (b, -a)\cdot(x, y) = \vec{v}^{\perp}\cdot(x, y) \end{cases}$$

$$u(x, y) \longrightarrow u(\xi, \eta)$$

$$u_x = \frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial x} = a\frac{\partial u}{\partial \xi} + b\frac{\partial u}{\partial \eta}$$

$$u_y = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial y} = b\frac{\partial u}{\partial \xi} - a\frac{\partial u}{\partial \eta}$$

$$\boxed{0 = a u_x + b u_y}$$

Hence $0 = a(a u_\xi + b u_\eta) + b(b u_\xi - a u_\eta)$

$$0 = a^2 u_\xi + \cancel{ab u_\eta} - \cancel{ba u_\eta} + b^2 u_\xi$$

Hence the transport eq. in (ξ, η) coordinates is simply $(a^2+b^2) u_\xi = 0 \quad a, b \neq 0$

$$\Longleftrightarrow \boxed{u_\xi = 0} \Longleftrightarrow \boxed{u(\xi, \eta) = f(\eta)}$$

where f is a function of 1 variable. This is the same as we got before because $\boxed{\eta = bx - ay}$.

So before $\boxed{u(x,y) = f(bx - ay)}$